



DMSPS: Dynamically mixed soft pseudo-label supervision for scribble-supervised medical image segmentation

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ABSTRACT

High performance of deep learning on medical image segmentation rely on large-scale pixel-level dense annotations, which poses a substantial burden on medical experts due to the laborious and time-consuming annotation process, particularly for 3D images. To reduce the labeling cost as well as maintain relatively satisfactory segmentation performance, weakly-supervised learning with sparse labels has attained increasing attentions. In this work, we present a scribble-based framework for medical image segmentation, called Dynamically Mixed Soft Pseudo-label Supervision (DMSPS). Concretely, we extend a backbone with an auxiliary decoder to form a dual-branch network to enhance the feature capture capability of the shared encoder. Considering that most pixels do not have labels and hard pseudo-labels tend to be over-confident to result in poor segmentation, we propose to use soft pseudo-labels generated by dynamically mixing the decoders' predictions as auxiliary supervision. To further enhance the model's performance, we adopt a two-stage approach where the sparse scribbles are expanded based on predictions with low uncertainties from the first-stage model, leading to more annotated pixels to train the second-stage model. Experiments on ACDC dataset for cardiac structure segmentation, WORD dataset for 3D abdominal organ segmentation and BraTS2020 dataset for 3D brain tumor segmentation showed that: (1) compared with the baseline, our method improved the average *DSC* from 50.46% to 89.51%, from 75.46% to 87.56% and from 52.61% to 76.53% on the three datasets, respectively; (2) DMSPS achieved better performance than five state-of-the-art scribble-supervised segmentation methods, and is generalizable to different segmentation backbones. The code is available online at: <https://github.com/HiLab-git/DMSPS>.

1. Introduction

Segmentation of multiple organs from medical images is an essential work in clinical diagnosis, treatment planning and follow-up of diseases, since it can provide 3D shape modeling and quantitative measurement of volume of different organs and their relationship (Luo et al., 2022b). As manual segmentation is time-consuming, labor-intensive and subject to expert knowledge of the operator, automatic segmentation is highly desired. Recently, Deep Neural Networks (DNNs) have made great progress in medical image segmentation due to their excellent ability to learn complex patterns and features directly from the data (Hesamian et al., 2019). However, their success relies heavily on large-scale high-quality pixel-level annotated images, where the time-consuming and expensive annotation process imposes a

large burden on radiologists and physicians, especially for multi-organ segmentation from 3D volumes (Luo et al., 2022b).

To reduce the annotation time and cost, several label-efficient methods have recently garnered attention, such as Semi-Supervised Learning (SSL) (Chen et al., 2021b; Zhong et al., 2023) and Weakly-Supervised Learning (WSL) (Lin et al., 2016; Zhao et al., 2024). SSL aims at leveraging a large amount of unlabeled images with a small set of labeled images to train the segmentation model. However, it still necessitates the utilization of highly accurate pixel-level annotations of some images, which is a cumbersome task for radiologists. In contrast, WSL requires sparser labels for training, such as image-level tags (Fu et al., 2023), bounding boxes (Wang and Xia, 2021), points (Zhang et al., 2023; Zhong and Wang, 2023) and scribbles (Gao et al., 2022), etc.,

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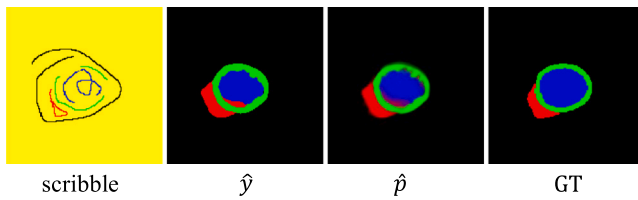


Fig. 1. Visual comparison between hard pseudo-labels $\hat{y}$ obtained by our previous work (Luo et al., 2022a) and soft pseudo-labels $\hat{y}$ of this work in learning from sparse annotations. In the last three pictures, red, green and blue channels denote the probability for right ventricle, myocardium and left ventricle, respectively. Best viewed in color.

leading to lower annotation cost while maintaining promising performance. As scribbles are convenient and efficient to annotate with more structural information, they are more suitable for non-convex shapes compared to other types of weak labels. Given these advantages, this work focuses on weakly supervised learning using scribble annotations.

Considering that only a small set of pixels are labeled in scribble annotations, effectively utilizing information from unlabeled ones is crucial for achieving optimal performance. Existing methods can be grouped into four categories: (1) Regularization losses that are model-agnostic and based on regularization energy (Tang et al., 2018a), such as Conditional Random Field (CRF) (Tang et al., 2018b; Obukhov et al., 2019), to encourage the model to produce results that align with the underlying structure of the image; (2) Consistency learning that encourages the network to produce consistent outputs for the same input under different perturbations, such as input augmentation (Zhang and Zhuang, 2022) and feature perturbation (Han et al., 2023); (3) Auxiliary task learning that introduces an auxiliary branch for a different representation of segmentation, such as a distance map (Gao et al., 2022) or an edge map (Chen and Hong, 2022), which imposes supplementary constraints on the model during training; and (4) Pseudo-label learning that jointly generates pseudo segmentation labels and trains the models iteratively (Lee and Jeong, 2020). Nevertheless, due to the lack of supervision, there is still a gap of performance between scribble-supervised learning and fully-supervised learning, leaving room for improvement.

Among the above methods, pseudo-label-based methods are promising since they can effectively leverage unlabeled regions by assigning pseudo-labels, thereby increasing the amount of labeled pixels available for training. However, they are limited by two problems. First, pseudo-labels generated by a model tend to be inherently biased towards that model's own predictions, which heavily limits the model's performance (Huo et al., 2021). Second, the commonly used hard pseudo-labels tend to be over-confident that may result in the risk of error propagation (Arazo et al., 2020; Xu et al., 2020). Some approaches (Lee and Jeong, 2020) try to use thresholds to carefully selected the hard pseudo-labels, while setting these thresholds too low can result in noisy pseudo-labels or too high may discard potentially useful information. In addition, most existing methods are designed for 2D slice segmentation with a single or few organs, and often require abundant and long scribbles for achieving high performance, and may perform poorly when segmenting multiple organs in 3D volumes or learning from more sparse scribbles. Therefore, more robust WSL segmentation methods that can handle multiple organs from 3D volumes and work effectively with weaker scribbles are desirable.

To deal with these problems, we present a simple yet efficient framework based on a Dual-Branch Network (DB-Net) with Dynamically Mixed Soft Pseudo-label Supervision (DMSPS). To mitigate the bias of pseudo-labels towards a single model, we propose mixing predictions randomly from two independent branches as pseudo-labels, where the model can take advantage of the strengths of each branch while diversifying the labels to minimize individual bias. Instead of relying on *hard* pseudo-labels, we propose using *soft* pseudo-labels as

auxiliary supervision during each iteration of network training. As shown in Fig. 1, in the hard pseudo-labels, a part of the left ventricle region is incorrectly taken as the right ventricle region. The label for the incorrect region is over-confident, which might mislead the segmentation model. In contrast, the soft pseudo-labels could indicate uncertainty among different classes and are more informative than one-hot hard labels (Lukasik et al., 2020). By using soft pseudo-labels, the model is supervised by a probability distribution at each pixel, which avoids the model being too rigid in its predictions, leading to higher noise-robust. In addition, to deal with the sparse supervision signals in scribble annotations, a two-stage training approach is adopted, where the first stage learns from the raw scribble annotations to generate high-quality pseudo-labels with uncertainty-filtering, and the confident pseudo-labels are used as expanded annotations in the second stage with less unannotated pixels to further enhance the model performance. The main contributions of this work are summarized as follows:

- (1) We propose a scribble-supervised framework that employs a dual-branch network with dynamically mixed pseudo-labels as auxiliary supervision, which mitigates the inherent bias of pseudo-labels from a single model to handle unlabeled pixels.
- (2) To alleviate the effect of noise in pseudo-labels, we propose soft pseudo-label supervision, which is more robust in learning from extremely sparse annotations.
- (3) To further improve the model performance, a two-stage training approach is adopted, where the uncertainty-based annotation expansion is introduced to handle unlabeled regions.

This work is a substantial extension of our previous work published in MICCAI 2022 (Luo et al., 2022a), where we introduced the dynamically mixed *hard* pseudo-labeling strategy that was only validated on a 2D segmentation task. The main extensions include: (1) Adopting *soft* pseudo-labels to replace hard pseudo-labels, which can better suppress the effect of over-confident noisy pseudo-labels; (2) Using a two-stage training method to further enhance the model's performance when learning from extremely sparse annotations; (3) Conducting experiments on two extra datasets for multi-organ segmentation and brain tumor segmentation in 3D volumes; and (4) Exploring the performance of different methods under different sparsity levels of scribbles and segmentation backbones. The experiments demonstrated that our method surpassed several state-of-the-art methods, and its performance was more robust when the scribble length was reduced by even 16 times.

2. Related works

2.1. Multi-organ medical image segmentation

Traditional medical image segmentation methods mainly use threshold, edge detection, region growing and other image processing-based methods for segmentation (Pham et al., 2000). However, their performance is poor in complex cases, as they require low-level features and manual selection and adjustment of parameters, making it difficult to adapt to different medical image segmentation tasks. Recently, deep learning methods such as Convolutional Neural Networks (CNNs) and Transformers have achieved high segmentation performance due to their automatic extraction of features and segmentation rules. The UNet model (Ronneberger et al., 2015) and its variants (Milletari et al., 2016; Zhou et al., 2019) are widely employed for medical image segmentation tasks. One of its most famous and powerful extension is nnUNet (Isensee et al., 2021) that includes a data augmentation pre-processing phase, a two-stage model optimization process, and an ensemble prediction mechanism to generate the final segmentation results. Transformer (Vaswani et al., 2017) based on self-attention mechanism has been used to handle long-range dependencies for multi-organs segmentation (Li et al., 2021). More recently, by integrating self-attention with the UNet structure, methods such as TransUNet (Chen

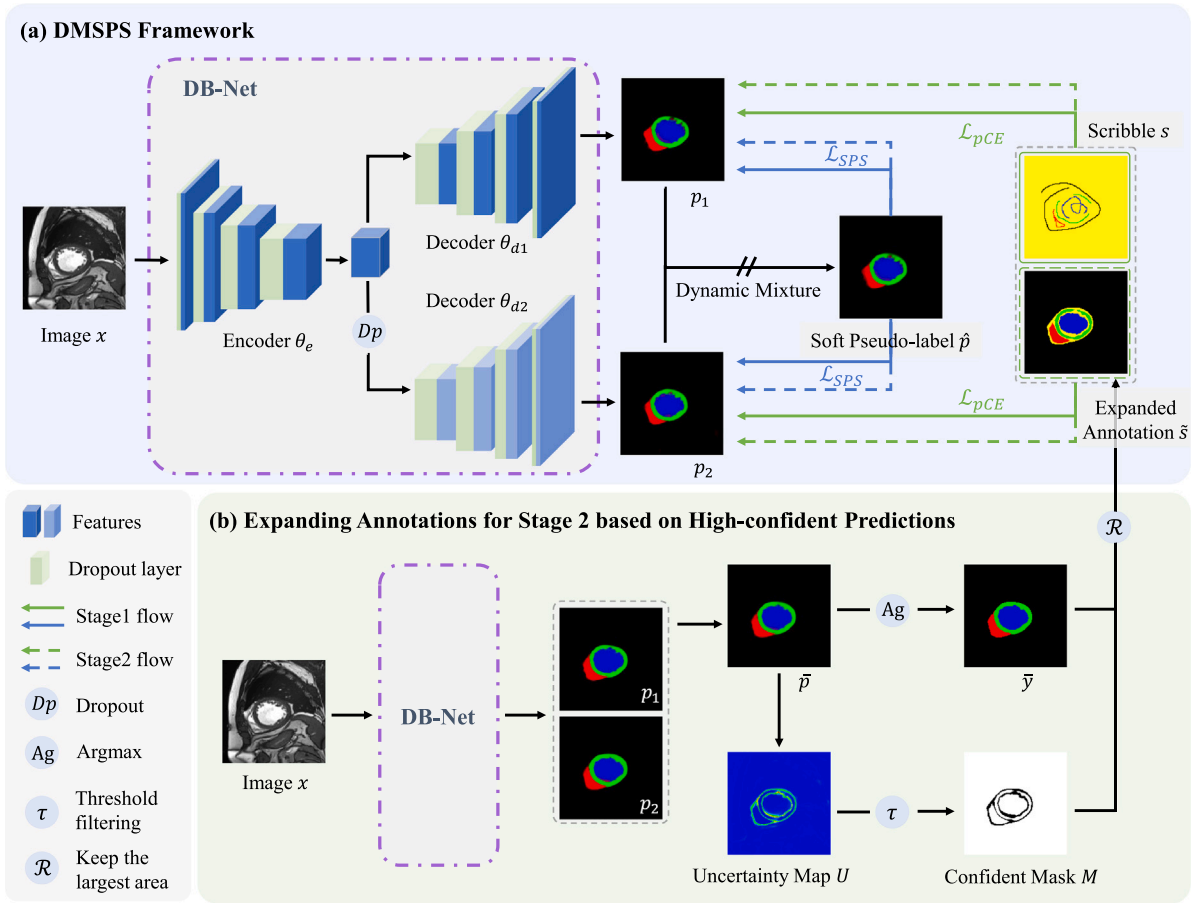


Fig. 2. Overview of our two-stage scribble-supervised segmentation framework through Dynamically Mixed Soft Pseudo-label Supervision (DMSPS). In the first stage, a Dual-Branch Network (DB-Net) is trained using raw scribble annotations s ($\mathcal{L}_{pCE}$) and the soft pseudo-label ($\mathcal{L}_{SPS}$) generated by dynamically mixing the two decoders' outputs. Reliable predictions from the first-stage model are used as expanded annotations $\tilde{s}$ to retrain the segmentation model in the second stage. Best viewed in color. Note that the yellow region in s and $\tilde{s}$ are unannotated pixels.

et al., 2021a) and UNETR (Hatamizadeh et al., 2022) have also been proposed for segmentation of multiple organs. However, the success of these methods heavily relies on dense pixel-level annotations for a training set, which are hard to obtain in practice, greatly limiting their clinical applicability.

2.2. Learning from scribbles for segmentation

Scribble annotation is one of the weak annotation forms, and it only requires the annotator to draw a few curves to mark a small part of each segmentation target and the background. The main challenge of scribble-supervised segmentation is lack of supervision of object structure and the uncertainty on the boundary. Recent methods are mainly based on regularization losses (Tang et al., 2018b,a; Obukhov et al., 2019), consistency learning (Liu et al., 2022; Zhang and Zhuang, 2022), auxiliary task learning (Gao et al., 2022; Chen and Hong, 2022) and pseudo-label learning (Can et al., 2018; Lee and Jeong, 2020; Zhou et al., 2023). For instance, Tang et al. (2018b) introduced a CRF-based regularization term that can be optimized together with the segmentation loss function. Liu et al. (2022) introduced transformation-consistency based on an uncertainty-aware mean teacher model. Gao et al. (2022) generated Geodesic distance maps for each category based on scribbles that were then utilized to construct a regression loss to supervise the model. Lee and Jeong (2020) utilized a warm-up strategy and label filtering in conjunction with pseudo-labeling to generate reliable labels for network training. Recently, some hybrid methods combining the above strategies have also been proposed. For example, Zhou et al. (2023) proposed a class-wise contrastive

regularization loss based on pseudo-labels, and Han et al. (2023) introduced a triple-branch network with different dilated convolution rates for feature-level perturbations and consistency regularization between each of them and their average. Despite the reduction of annotation cost by learning from scribbles, the performance of these methods is still inferior to that achieved by training with dense annotations. In addition, most existing methods require long scribbles to get satisfactory results, and their performance will decrease largely when the scribbles become shorter and sparser.

2.3. Pseudo-label learning

Pseudo-labeling is widely used in annotation-efficient tasks like WSL (Liang et al., 2022; Zhou et al., 2022) and SSL (Zou et al., 2020; Basak and Yin, 2023). The core idea is to deal with unannotated pixels by generating pseudo segmentation labels for model training. Pseudo-label generation methods can be roughly divided into three categories: (1) Using traditional segmentation methods for label propagation, which assigns labels to unlabeled pixels based on the similarity to labeled ones. Roth et al. (2021) used random walker to obtain an initial segmentation based on a small set of annotated pixels, and ScribbleSup (Lin et al., 2016) used a graph-based approach to propagate labels from scribbles to unlabeled pixels. (2) Using predictions from a deep learning model as pseudo-labels for itself. For instance, Can et al. (2018) and Rajchl et al. (2017) employed a recursive training framework with alternative network parameter update and CRF-based relabeling of training data. (3) Using a different network or branch to obtain pseudo labels to supervise a model. For example, Chen

et al. (2021b) introduced Cross Pseudo Supervision (CPS) between two networks for semi-supervised learning. As pseudo-labels generated from weak annotations tend to be noisy, various techniques such as labeling filters (Lee and Jeong, 2020), affinity matrices (Ru et al., 2022) and uncertainty guidance (Wang et al., 2021) have been used for refinement. However, they either require setting complex task-specific hyper-parameters or additional computational resources for optimization. What is more, most of them use hard pseudo-labels that augment the impact of noise (Xu et al., 2020). Instead, we propose to use soft pseudo-labels that are more noise-robust for training, suppressing the effect of potentially inaccurate ones.

3. Method

The proposed DMSPS framework for scribble-supervised medical image segmentation is depicted in Fig. 2. A dual-branch network with a shared encoder and two slightly different decoders is proposed to learn from scribble annotations. The two branches' outputs are dynamically mixed to generate soft pseudo-labels to assist the network training. To further enhance the model's ability to deal with sparse annotations, we introduce a two-stage training strategy, where the sparse annotations are propagated to larger regions based on low-uncertainty predictions from the first stage, leading to more annotated pixels for training the second-stage model with the same architecture and losses used in the first stage.

To describe this work precisely, we first introduce several mathematical symbols. Let $X = \{x_n\}$, $S = \{s_n\}$ ($n = 1, \dots, N$) be the training images and the corresponding scribble annotations, respectively. N is the number of training images. Let $\Omega = \Omega_L \cup \Omega_U$ denote the whole pixel set in an image x , where Ω_L and Ω_U refer to the labeled and unlabeled regions, respectively. Let C represent the number of categories for segmentation. Note that the partial annotation s of an image has $C + 1$ classes, where the extra class means unlabeled pixels.

3.1. Dual-branch network

The proposed Dual-Branch Network (DB-Net) consists of a shared encoder (θ_e) for feature extraction and two independent decoders (θ_{d1} , θ_{d2}) for segmentation and supplementary training. We constructed the network by embedding a perturbed decoder into the general U-Net (Ronneberger et al., 2015) and 3D U-Net (Çiçek et al., 2016) for 2D and 3D segmentation tasks respectively, where dropout is used on the bottleneck's output feature before sending it into the auxiliary decoder, as shown in Fig. 2(a). Concretely, the encoder is constructed with multiple cascaded convolution-downsampling units. Each unit comprises two convolution blocks, a dropout layer and a downsampling layer. Within each convolution block, there is a convolution layer, a batch normalization layer for 2D or instance normalization layer for 3D, and a ReLU activation function layer. Similarly, each decoder consists of multiple cascaded convolution-upsampling units, where each unit has two convolution blocks and an upsampling layer. The structure of the convolution blocks in the decoder corresponds to those in the encoder. It should be noted that the dropout within the network architecture is used to prevent over-fitting, while the dropout in the bottleneck is employed to introduce additional perturbation.

Note that the two decoders have different predictions and decision boundaries, so that they can supervise each other and avoid the bias towards a single prediction branch. In addition, the dual-branch network not only improves the feature extraction ability of the encoder (Ouali et al., 2020; Wu et al., 2021), but also reduces the parameter number and computational cost compared with using two segmentation networks (Chen et al., 2021b). Despite that dual-branch networks have been introduced for mixed supervision (Liu et al., 2023) and noisy label learning (Yang et al., 2022), there are significant differences in the learning task and supervision strategies. First, in terms of annotation requirement, Liu et al. (2023) learns from a fully

annotated subset combined with a sparsely annotated subset, and Yang et al. (2022) learned from a set of images with clean labels and another set with noisy labels. Both of them require dense annotations of some images, while our method assumes that only sparse scribble annotations are available. Second, they use unidirectional supervision, i.e., one branch learning from full annotations supervises the other branch on sparse/noise annotations. Therefore, these methods cannot be used in the scenario of our work where all the training images do not have full annotations. To deal with this problem, our method uses a different supervision strategy, i.e., mutual supervision based dynamically mixed soft pseudo-labels, which provides effective pseudo supervision signals for unannotated pixels and avoids the model being corrupted by bias and noise of a single prediction branch.

3.2. Dynamically mixed soft pseudo label

Due to the insufficient supervision signal in sparse annotations, models trained with only scribbles cannot achieve promising results. To solve this dilemma, we propose to exploit the decoders' outputs by randomly mixing them to generate pseudo-labels to further boost the model training. Rather than utilizing hard pseudo-labels that tend to be over-confident (Arazo et al., 2020) on incorrect pseudo supervision, we propose to use soft pseudo-labels to provide probabilistic supervision that is able to encode inter-class similarity and less noisy than one-hot hard labels (Xu et al., 2020). Let p_1 and p_2 denote the softmax outputs of the two decoders, respectively. The dynamically mixed soft pseudo-label $\hat{p}$ is defined as:

$$\hat{p} = \alpha \times p_1 + (1.0 - \alpha) \times p_2 \quad (1)$$

where $\alpha \sim U(0, 1)$ is the mixture coefficient, and is randomly sampled from a uniform distribution in the range of $[0, 1]$ at each iteration. Note that $\hat{p}$ is a dynamic ensemble of p_1 and p_2 on-the-fly controlled by the random coefficient α during training. For example, when $\alpha = 0.0$ and $\alpha = 1.0$, $\hat{p}$ becomes p_2 and p_1 , respectively. When $\alpha = 0.5$, $\hat{p}$ is an average of p_1 and p_2 . The dynamic mixture enhances the diversity of pseudo-labels and mitigates the inherent drawback of single-branch pseudo-labeling, i.e., over-fitting to the bias of the model (Huo et al., 2021). Then, we use the soft pseudo-label $\hat{p}$ to supervise the two decoders respectively. The Soft Pseudo-label Supervision (SPS) loss is defined as:

$$\mathcal{L}_{SPS} = \frac{1}{2} \times [L(p_1 \parallel \hat{p}) + L(p_2 \parallel \hat{p})] \quad (2)$$

where $L(\cdot \parallel \cdot)$ refers to the segmentation loss function. In this paper, we adopt the cross-entropy loss, and it can also be replaced by the Kullback–Leibler divergence, the Mean Squared Error (MSE) loss and others.

Note that $\hat{p}$ is detached before being sent to Eq. (2), which cuts off the gradient of $\mathcal{L}_{SPS}$ to $\hat{p}$ for pseudo label supervision rather than enforcing consistency directly (Zhang et al., 2018; Ouali et al., 2020; Wu et al., 2021). In addition, we do not use argmax of $\hat{p}$ in Eq. (2) as that will remove the uncertainty information in the pseudo-label and enlarge the effect of potentially wrong pseudo-labels. In contrast, the soft pseudo-label can encode the similarity between different classes and represent the uncertainty in the pseudo-label, which reduces the effect of noise.

3.3. Overall loss function

For annotated pixels in the scribbles, we apply the partial Cross-Entropy (pCE) loss to each of the decoders, which ignores the unlabeled pixels during gradient back-propagation and only takes the labeled ones

into consideration:

$$\mathcal{L}_{pCE} = -\frac{1}{2|\Omega_L|} \sum_{i \in \Omega_L} \sum_c (s_i^c \log(p_{1,i}^c) + s_i^c \log(p_{2,i}^c)) \quad (3)$$

where $p_{1,i}^c$ and $p_{2,i}^c$ denote the probability of pixel i belonging to class c predicted by the two branches, respectively. Ω_L is the set of labeled pixels in a scribble annotation s .

Finally, the overall loss function for training is defined as:

$$\mathcal{L} = \mathcal{L}_{pCE} + \lambda \mathcal{L}_{SPS} \quad (4)$$

where λ is a weight factor to balance the supervision from scribbles and soft pseudo-labels.

3.4. Second-stage training with expanded annotations

Though Eq. (4) uses soft pseudo-labels that are more robust to noisy pseudo-labels than hard ones, the supervision loss $\mathcal{L}_{pCE}$ is only calculated on a very small set of annotated pixels, which has a risk of over-fitting and can hardly ensure that the predictions for unannotated pixels are well-calibrated. To deal with this problem, we propose a two-stage approach by expanding the scribbles based on high-confidence predictions from the first-stage model trained with Eq. (4), and retrain the segmentation model with less unannotated pixels.

As illustrated in Fig. 2(b), we take an entropy-based uncertainty to exclude low-confident predictions. For each training image, we use the best model trained from the first stage (using the raw scribble annotations for training) for inference. The probability prediction maps p_1 and p_2 are averaged, i.e., α is set to its expectation of 0.5, and the result is denoted as $\bar{p}$. The pseudo label based on an argmax of $\bar{p}$ is denoted as $\bar{y}$. The uncertainty map for average prediction is denoted as U , and an element of it is defined as:

$$U(i) = -\frac{1}{\log(C)} \sum_{c=0}^{C-1} \bar{p}_i^c \log(\bar{p}_i^c) \quad (5)$$

where $\log(C)$ normalizes the uncertainty to the range of (0,1), eliminating the impact of the number of categories. Then we introduce a threshold τ to obtain a mask M for reliable regions:

$$M = U < \tau \quad (6)$$

Then, we take non-zero region of M as expanded annotation set Ω_M , and the corresponding values in $\bar{y}$ as their labels. To distinguish the expanded annotation from the original scribble annotation s , we use $\tilde{s}$ to denote the final uncertainty-filtered high-confident expanded annotation for the second stage:

$$\tilde{s} = \mathcal{R}(\bar{y} \odot M) \quad (7)$$

where $\mathcal{R}$ denotes morphological operation to retain the largest connected component of each class to remove noise. With the expanded annotation $\tilde{s}$, we retrain the segmentation model by defining the partial CE loss on Ω_M , and Eq. (3) is modified to:

$$\mathcal{L}'_{pCE} = -\frac{1}{2|\Omega_M|} \sum_{i \in \Omega_M} \sum_c (\tilde{s}_i^c \log(p_{1,i}^c) + \tilde{s}_i^c \log(p_{2,i}^c)) \quad (8)$$

The second-stage model is initialized by the optimal model from the first stage, and the loss function is:

$$\mathcal{L}' = \mathcal{L}'_{pCE} + \lambda \mathcal{L}_{SPS} \quad (9)$$

4. Experiments and analysis

4.1. Dataset

We evaluated our method and compared it with several state-of-the-art methods on three public datasets: the ACDC (Automated Cardiac Diagnosis Challenge) dataset for cardiac structure segmentation from

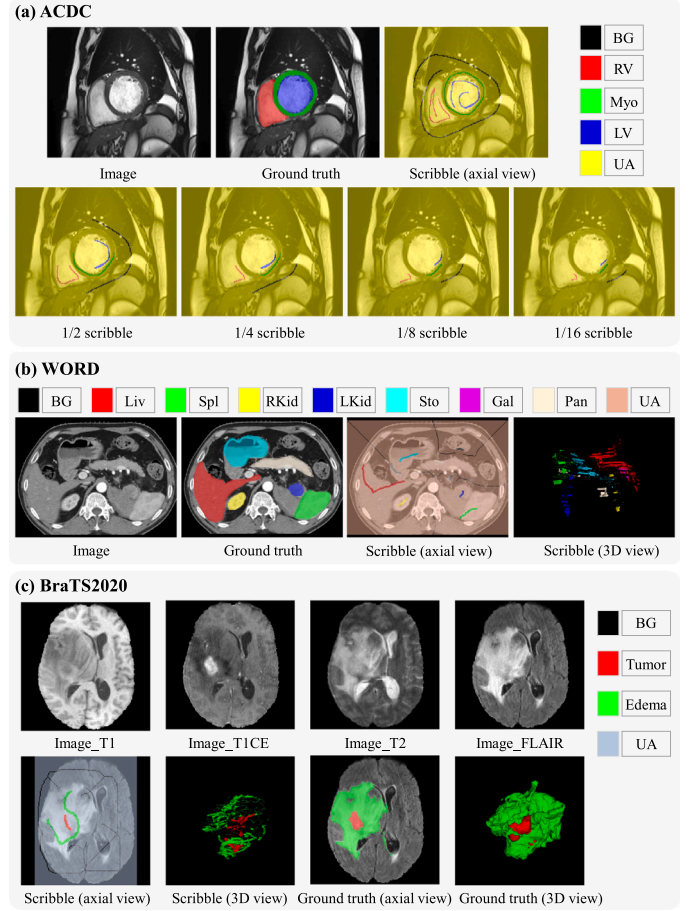


Fig. 3. Examples of scribble annotations on ACDC, WORD and BraTS2020 datasets. In (a), BG, RV, Myo, LV and UA represent the background, right ventricle, myocardium, left ventricle and unannotated pixels, respectively. The second line presents the scribbles with reduced length. In (b), Liv, Spl, RKid, LKid, Sto, Gal and Pan represent the liver, spleen, left kidney, right kidney, stomach, gallbladder and pancreas, respectively. Best viewed in color.

2D slices, WORD dataset for multi-organ segmentation from 3D volumes, and BraTS2020 dataset for more complex tumor segmentation task with irregular boundaries.

The ACDC dataset (Bernard et al., 2018) comprises 200 short-axis cine-Magnetic Resonance Imaging (MRI) scans from 100 patients obtained by different magnetic strengths and temporal resolutions. Each scan contains 6–18 slices with varying dimensions, ranging from 154–276 in height and 174–384 in width. For each patient, dense annotations of the Right Ventricle (RV), Myocardium (MYO) and Left Ventricle (LV) were manually provided for both the end-diastolic and end-systolic phase. Following Valvano et al. (2021), we divided the data at the patient level into 140 cases from 70 patients, 30 cases from 15 patients and 30 cases from the other 15 patients for training, validation and testing, respectively. Considering the large inter-slice spacing (mostly 10 mm), we used 2D networks for slice-by-slice segmentation, and stacked the results into a volume for 3D evaluation. We used the scribble annotations provided by Valvano et al. (2021) for weakly supervised learning. As shown in Fig. 3(a), the original scribbles are quite long for each class. To investigate the effectiveness of different methods with fewer scribbles, we cut the length of each scribble to 1/2, 1/4, 1/8 and 1/16 respectively, leading to different degrees of sparse annotations.

The WORD dataset (Luo et al., 2022b) contains 150 abdominal CT volumes collected from 150 patients before radiation therapy at a single center. Each CT volume consists of 159–330 slices of 512×512 pixels,

with an in-plane resolution of $0.976 \text{ mm} \times 0.976 \text{ mm}$ and inter-slice spacing of $2.5\text{--}3.0 \text{ mm}$. We aimed to segment seven organs: the liver, spleen, left kidney, right kidney, stomach, gallbladder and pancreas. As shown in Fig. 3(b), the scribble annotations for foreground organs and background were provided in the axial view by Luo et al. (2022b), and they were generated by a script to imitate human drawing. The dataset was split into 100, 20 and 30 volumes for training, validation and testing, respectively.

The BraTS2020 dataset (Menze et al., 2014; Bakas et al., 2017, 2018) contains 369 MRI volumes with 4 modalities: T1, T1ce, T2 and FLAIR (Fluid Attenuated Inversion Recovery). They have a uniform resolution of 1.0 mm^3 with a size of $240 \times 240 \times 155$. We aimed to segment two foreground classes: the tumor core and the peritumoral edema. The dataset was randomly split into 70%, 15% and 15% for training, validation and testing, respectively. Following the practice on WORD, the scribble annotations were generated in the axial view by a script to imitate human drawing (Luo, 2021), as shown in Fig. 3(c).

4.2. Implementations details and evaluation metrics

2D segmentation: For preprocessing of the ACDC dataset, intensity values in each slice were rescaled to $[0,1]$ based on the minimal value and the 99-th percentile. The UNet (Ronneberger et al., 2015) was used as the backbone, and we extended it to a dual-branch network by adding an auxiliary decoder, with a dropout layer (ratio=0.5) in the bottleneck before the auxiliary decoder to introduce perturbations. For network training, random rotation and flipping were used for data augmentation, and the augmented images were resized to 256×256 as the input. The batch size and total iteration number were set to 24 and 30k (≈ 527 epochs), respectively. For testing, we produced predictions slice by slice and then stacked them into a 3D volume.

3D segmentation: On the WORD dataset, we applied a fixed window/level of 400/50 to selectively focus on the abdominal organs, and then normalized the Hounsfield Unit values to the range of $[0,1]$. For the BraTS2020 dataset, images were first cropped based on the bounding box of non-zero regions, and the intensity of each modality was rescaled to $[0,1]$ based on the minimal value and the 99-th percentile. The four modalities were concatenated to be a 4-channel input for the network. Similarly to 2D segmentation, we employed the 3D UNet (Çiçek et al., 2016) as the backbone, and extended it to a 3D dual-branch network. For both WORD and BraTS2020 dataset, data augmentation was performed solely through random cropping with a patch size of $80 \times 96 \times 96$ during training. The batch size was set to 1 due to the memory limitation, with a maximum of 60k iterations (≈ 600 epochs). The final segmentation results were obtained by using a sliding window strategy for both WORD and BraTS2020 datasets.

All methods were implemented by PyTorch 1.11 (Paszke et al., 2019) and PyMIC (Wang et al., 2023), using an NVIDIA GTX2080TI GPUs with 11 GB memory. Note that due to changes in hardware, the numerical results reported in this paper may differ from the MICCAI version (Luo et al., 2022a), but the overall trends and conclusions from them are consistent. The Stochastic Gradient Descent (SGD) optimizer with initial learning rate (η_0) of 0.01, momentum of 0.9 and weight decay of 10^{-4} was used. The poly learning rate scheduler was employed, and the learning rate for t -th iteration was $\eta_t = (1.0 - t/t_{max})^{0.9} \eta_0$, where t_{max} refers to the maximum iteration number. According to the best performance on the validation set, λ was set to 8.0 for all three datasets, and τ was 0.1, 0.3 and 0.4 for the ACDC, WORD, BraTS2020 datasets respectively. The second-stage model was initialized from the best first-stage model, and early stopping (tolerance 10k) was used to save time. We used the primary decoder's output at inference time and did not use any post-processing method for both 2D and 3D segmentation tasks. Two common metrics were used for quantitative evaluation: (1) The 3D Dice Similarity Coefficient (DSC) was used to evaluate the pixel-wise overlap between the ground truth and prediction, (2) 95% Hausdorff Distance (HD_{95}) was adopted to measure the surface distance between the ground truth and prediction in 3D space.

4.3. Comparison with existing weakly-supervised methods

Our DMSPS was compared with five state-of-the-art weakly-supervised learning methods: (1) Uncertainty-aware Self-ensembling and Transformation-consistent Model (USTM) (Liu et al., 2022) that encourages the teacher and student outputs to be consistent under different perturbations with uncertainty estimation to suppress unreliable supervision from the teacher; (2) Scribble2Label (S2L) (Lee and Jeong, 2020) that combines pseudo-labeling and label filtering to generate reliable pseudo-labels for supervision; (3) Entropy Minimization (EM) (Grandvalet and Bengio, 2004) that minimizes the entropy of predictions on unannotated pixels; (4) CycleMix (Zhang and Zhuang, 2022) that uses mix augmentation and cycle consistency for regularization; (5) Gated CRF Loss (GCL) (Obukhov et al., 2019) that incorporates the gating function with CRF loss to regularize the prediction. They were also compared with the lower bound using only pCE loss for supervision, as well as the upper bound using dense annotations for fully supervised learning (FullySup). All these methods were implemented with the same backbone network for each dataset.

4.3.1. Results on ACDC dataset

Table 1 shows quantitative results on the ACDC dataset with the raw scribbles provided by Valvano et al. (2021). It can be seen that all the existing weakly supervised methods had a large performance improvement from the baseline of pCE. Among the existing methods, GCL (Obukhov et al., 2019) obtained the best performance, with average DSC and HD_{95} being 87.89% and 4.37 mm, respectively. In contrast, our method in the first stage outperformed GCL, with the average DSC and HD_{95} being 88.94% and 4.22 mm, respectively. Our method in the second stage further improved the average DSC to 89.51%, and reduced the average HD_{95} to 3.72 mm, respectively. Our method significantly outperformed GCL on Myo and LV in term of average DSC (p -value < 0.05 based on paired t -test). In addition, the performance of our method is very close to that of the FullySup upper bound on RV (average DSC 88.56% vs 88.71%) and LV (average DSC 93.22% vs 93.27%), showing its ability to maintain high performance with sparse annotations.

For simple segmentation tasks such as ACDC, traditional segmentation methods such as GraphCuts (Boykov and Jolly, 2001) and Random Walker (Grady, 2006) might be used for segmentation as well. For a comparison, we directly applied these methods for inference on the test set with scribbles, and they are denoted as RW-Test and GraphCuts-Test, respectively. In addition, we also used Random Walker and GraphCuts to obtain segmentation results of the training images, the output of which was used to train a UNet for segmentation. They are denoted as RW + UNet and GraphCuts + UNet, respectively in Table 1. The results in Table 1 show that all these four methods outperformed the lower bound (UNet + only pCE). GraphCuts + UNet even outperformed USTM in terms of average DSC (80.81% vs 79.92%) and HD_{95} (7.41 mm vs 85.84 mm). However, these traditional methods are far inferior to our method in terms of both DSC and HD_{95} . For example, the first-stage model of our method already obtained an average DSC of 88.94% and HD_{95} of 4.22 mm, respectively.

To further investigate the performance of different methods with fewer scribbles, we experimented with the ACDC dataset by reducing the length of the raw scribbles to 1/2, 1/4, 1/8 and 1/16, respectively. The second section in Table 1 shows the results of different methods when the scribble length is reduced to 1/4. All methods experienced some degree of performance degradation with shorter scribbles. For example, the average DSC of the best existing method GCL decreased by 3.14 percentage points to 84.75%. In contrast, the corresponding value of our method only dropped from 89.51% to 87.59%.

The average DSC values of different methods with the decrease of scribble length are shown in Fig. 4. It can be observed that our method only has a slight performance decrease when the scribbles become sparser, and it consistently outperforms the existing weakly

Table 1

Quantitative comparison between our method DMSPS and existing weakly supervised methods under different annotation settings on ACDC testing set. “RW” means the traditional segmentation method “Random Walker”. The best values are highlighted in bold.

Annotation	Method	RV		Myo		LV		Mean	
		$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$	$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$	$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$	$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$
Full mask	FS (upper bound)	88.71 $\pm$ 7.91	5.10 $\pm$ 4.29	88.95 $\pm$ 4.30	3.17 $\pm$ 3.10	93.27 $\pm$ 7.59	3.68 $\pm$ 4.75	90.31 $\pm$ 6.60	3.98 $\pm$ 4.05
	pCE (lower bound)	46.53 $\pm$ 14.15	194.93 $\pm$ 31.71	46.19 $\pm$ 11.17	171.28 $\pm$ 20.91	58.65 $\pm$ 20.00	169.98 $\pm$ 21.44	50.46 $\pm$ 15.11	178.73 $\pm$ 24.68
Raw scribbles	USTM (Liu et al., 2022)	84.91 $\pm$ 7.77	37.87 $\pm$ 51.66	74.76 $\pm$ 8.26	111.23 $\pm$ 55.19	80.10 $\pm$ 15.60	107.36 $\pm$ 65.31	79.92 $\pm$ 10.54	85.48 $\pm$ 57.39
	S2L (Lee and Jeong, 2020)	84.96 $\pm$ 8.05	44.43 $\pm$ 60.36	79.27 $\pm$ 6.44	75.60 $\pm$ 54.44	82.92 $\pm$ 12.28	128.67 $\pm$ 78.90	82.38 $\pm$ 8.92	82.90 $\pm$ 64.57
	EM (Grandvalet and Bengio, 2004)	85.90 $\pm$ 8.36	17.61 $\pm$ 33.26	79.22 $\pm$ 6.68	59.94 $\pm$ 54.75	87.53 $\pm$ 9.18	51.18 $\pm$ 55.55	84.22 $\pm$ 8.07	42.91 $\pm$ 47.85
	CycleMix (Zhang and Zhuang, 2022)	87.12 $\pm$ 6.93	10.30 $\pm$ 18.25	82.12 $\pm$ 5.68	37.35 $\pm$ 40.04	89.10 $\pm$ 8.90	43.21 $\pm$ 50.31	86.11 $\pm$ 7.17	30.29 $\pm$ 36.20
	GCL (Obukhov et al., 2019)	88.23 $\pm$ 6.69	5.28 $\pm$ 3.68	83.20 $\pm$ 5.21	4.21 $\pm$ 2.62	92.23 $\pm$ 7.63	3.63 $\pm$ 4.70	87.89 $\pm$ 6.51	4.37 $\pm$ 3.67
	DMSPS-stage1	88.26 $\pm$ 6.76	5.09 $\pm$ 3.74	85.62 $\pm$ 5.40*	3.78 $\pm$ 3.72	92.95 $\pm$ 5.84	3.78 $\pm$ 5.35	88.94 $\pm$ 6.00*	4.22 $\pm$ 4.27
	DMSPS-stage2	88.58 $\pm$ 6.52	4.62 $\pm$ 3.70	86.74 $\pm$ 4.64*	2.96 $\pm$ 2.40*	93.22 $\pm$ 5.77	3.57 $\pm$ 4.76	89.51 $\pm$ 5.64*	3.72 $\pm$ 3.62*
	RW-Test (Grady, 2006)	85.08 $\pm$ 8.15	4.92 $\pm$ 2.24	67.50 $\pm$ 7.63	9.29 $\pm$ 3.55	82.83 $\pm$ 6.94	7.57 $\pm$ 3.85	78.47 $\pm$ 7.57	7.26 $\pm$ 3.21
	RW + UNet	83.43 $\pm$ 6.67	6.70 $\pm$ 3.96	67.90 $\pm$ 9.57	10.31 $\pm$ 6.79	82.16 $\pm$ 10.01	8.70 $\pm$ 5.58	77.83 $\pm$ 8.75	8.57 $\pm$ 5.44
	GraphCuts-Test (Boykov and Jolly, 2001)	80.18 $\pm$ 12.03	7.91 $\pm$ 4.34	74.32 $\pm$ 6.20	7.36 $\pm$ 1.68	84.78 $\pm$ 7.11	5.45 $\pm$ 2.14	79.76 $\pm$ 8.45	6.91 $\pm$ 2.72
	GraphCuts + UNet	81.12 $\pm$ 10.00	8.11 $\pm$ 4.64	74.83 $\pm$ 5.93	7.58 $\pm$ 3.90	86.49 $\pm$ 8.44	6.73 $\pm$ 5.99	80.81 $\pm$ 8.13	7.47 $\pm$ 4.84
1/4 scribbles	pCE (lower bound)	26.96 $\pm$ 11.54	189.32 $\pm$ 25.71	25.41 $\pm$ 5.19	180.89 $\pm$ 20.20	40.85 $\pm$ 16.17	190.34 $\pm$ 21.48	31.07 $\pm$ 10.96	186.85 $\pm$ 22.46
	USTM (Liu et al., 2022)	71.66 $\pm$ 14.02	102.52 $\pm$ 68.45	60.72 $\pm$ 13.04	148.25 $\pm$ 22.52	27.84 $\pm$ 18.88	186.23 $\pm$ 21.51	53.40 $\pm$ 15.31	145.67 $\pm$ 37.49
	S2L (Lee and Jeong, 2020)	75.25 $\pm$ 13.14	26.49 $\pm$ 36.78	79.43 $\pm$ 6.39	33.99 $\pm$ 43.34	76.87 $\pm$ 17.46	121.28 $\pm$ 63.75	77.19 $\pm$ 12.33	60.59 $\pm$ 47.96
	EM (Grandvalet and Bengio, 2004)	76.63 $\pm$ 12.86	19.86 $\pm$ 25.13	79.25 $\pm$ 7.51	48.90 $\pm$ 45.30	87.46 $\pm$ 10.45	56.60 $\pm$ 59.13	81.11 $\pm$ 10.28	41.79 $\pm$ 43.19
	CycleMix (Zhang and Zhuang, 2022)	81.93 $\pm$ 11.60	11.06 $\pm$ 11.72	82.44 $\pm$ 5.99	21.16 $\pm$ 28.46	87.41 $\pm$ 11.28	33.97 $\pm$ 44.72	83.93 $\pm$ 9.62	22.06 $\pm$ 28.30
	GCL (Obukhov et al., 2019)	84.63 $\pm$ 11.25	6.89 $\pm$ 4.57	80.89 $\pm$ 4.94	4.84 $\pm$ 2.42	88.73 $\pm$ 7.90	7.18 $\pm$ 10.69	84.75 $\pm$ 8.03	6.31 $\pm$ 5.90
	DMSPS-stage1	84.71 $\pm$ 10.88	6.89 $\pm$ 3.85	84.26 $\pm$ 5.57*	3.85 $\pm$ 2.59*	91.37 $\pm$ 9.31*	5.61 $\pm$ 7.97	86.78 $\pm$ 8.59*	5.45 $\pm$ 5.05*
	DMSPS-stage2	85.71 $\pm$ 10.50	6.49 $\pm$ 4.51	85.71 $\pm$ 10.50*	3.39 $\pm$ 2.36*	91.96 $\pm$ 8.54*	4.29 $\pm$ 5.67	87.59 $\pm$ 8.16*	4.72 $\pm$ 4.18*

* Denotes p -value < 0.05 (paired t -test) when comparing with the second place method GCL (Obukhov et al., 2019).

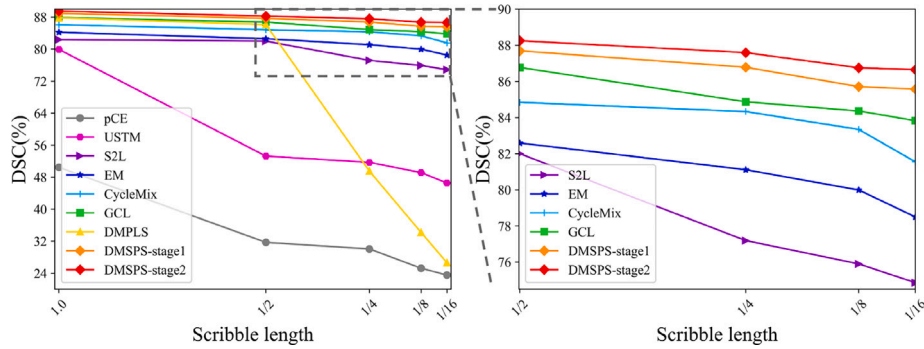


Fig. 4. Segmentation performance of different methods supervised by scribbles with different length. Best viewed in color.

supervised methods, showing the stability of our method. In contrast, the performance of the other methods such as pCE and USTM (Liu et al., 2022) dropped largely with reduced scribbles. Note that our previous work DMPLS (Luo et al., 2022a) that uses hard pseudo-labels for supervision outperformed existing methods when the raw scribbles were used. However, its performance dropped largely when the scribbles were reduced to 1/4 and shorter, mainly due to that the hard pseudo labels are more noisy and unreliable when the annotations are extremely sparse.

Fig. 5 shows a visual comparison of results obtained by different methods under different levels of scribble length. Note that with the decrease of scribble length, the performance of pCE dropped largely. EM (Grandvalet and Bengio, 2004), CycleMix (Zhang and Zhuang, 2022) and GCL (Obukhov et al., 2019) also obtained more mis-segmentation when the scribble length was reduced to 1/4. In contrast, our method was more stable under different length of scribbles.

4.3.2. Results on WORD dataset

We further validated our method on the more challenging WORD dataset for 3D segmentation of multiple organs. Quantitative evaluation results of different methods in terms of DSC and HD_{95} on the WORD dataset are presented in Table 2. As shown, pCE achieved an average DSC of 75.46%. USTM (Liu et al., 2022) and S2L (Lee and Jeong, 2020) obtained an average DSC of 78.85% and 78.39%, respectively. GCL (Obukhov et al., 2019) achieved the highest DSC (85.12%) among the existing methods. It can be found that our DMSPS in the first stage outperformed GCL (Obukhov et al., 2019), and obtained an average DSC of 86.79% and HD_{95} of 8.85 mm, respectively. After training

in the second stage, our method further improved the average DSC to 87.56% and reduced the average HD_{95} to 6.27 mm, respectively. Moreover, our method exhibits a significant improvement over the best existing method GCL (Obukhov et al., 2019) in terms of DSC and HD_{95} on most organs, especially complex or small-sized organs such as the stomach, gallbladder and pancreas.

The visual comparison in Fig. 6 shows that pCE obtained a lot of false positives, and EM (Grandvalet and Bengio, 2004) and CycleMix (Zhang and Zhuang, 2022) had some noisy segmentation results in the 3D space. Compared with the existing methods, our DMSPS model achieved more accurate segmentation results with fewer over-/under-segmentation regions.

4.3.3. Results on BraTS2020 dataset

We further applied our method to the more challenging task of brain tumor segmentation on the BraTS2020 dataset. Fig. 7 shows a visual comparison with existing methods. Note that no post-processing was used, and most existing methods obtained a lot of noise in their predictions, while our method attained better results already in the first stage. The second stage further makes the prediction at local regions closer to the ground truth. Quantitative comparison with existing methods in Table 3 shows that pCE achieved an average DSC of 53.61% and average HD_{95} of 70.98 mm as the lower bound. GCL achieved the highest DSC of 72.55% among existing methods. Our first-stage method significantly outperformed GCL, with an average DSC of 74.28% and HD_{95} of 10.75 mm. Performance was further markedly improved by our second-stage model with an average DSC of 76.53% and HD_{95} of 7.85 mm, respectively.

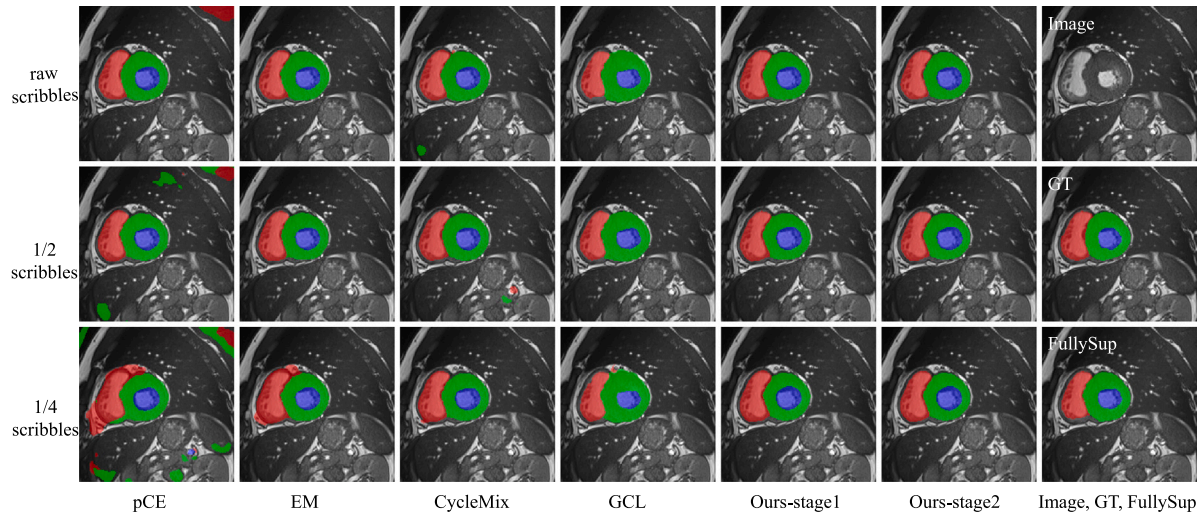


Fig. 5. Visual comparison between different weakly supervised methods under different levels of scribble length on the ACDC dataset. Best viewed in color.

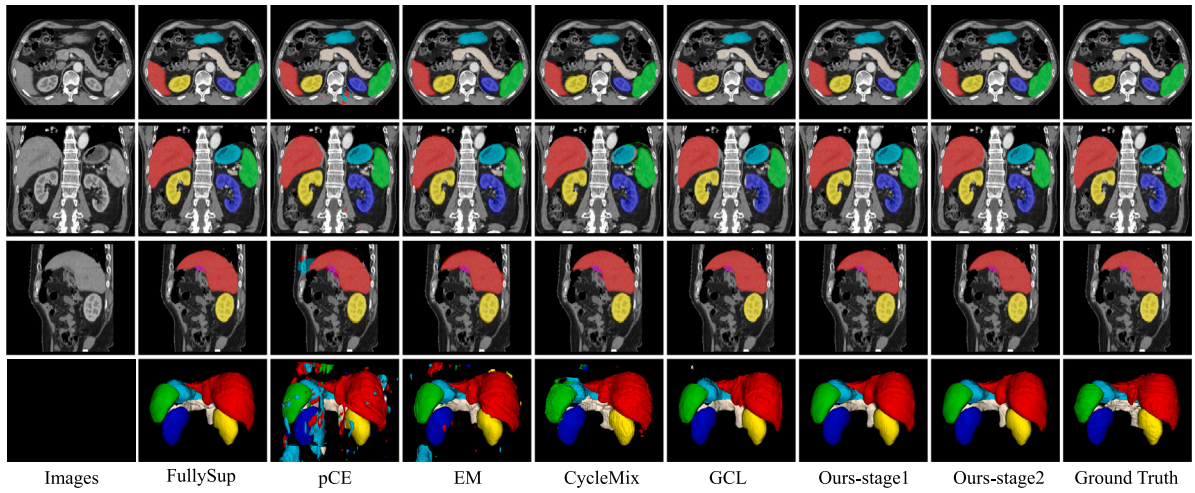


Fig. 6. Visual comparison between our method and the other weakly supervised methods on the WORD dataset. The 1st, 2nd, 3rd, and 4th rows display the visualizations of axial, coronal, sagittal, and 3D views, respectively. Best viewed in color.

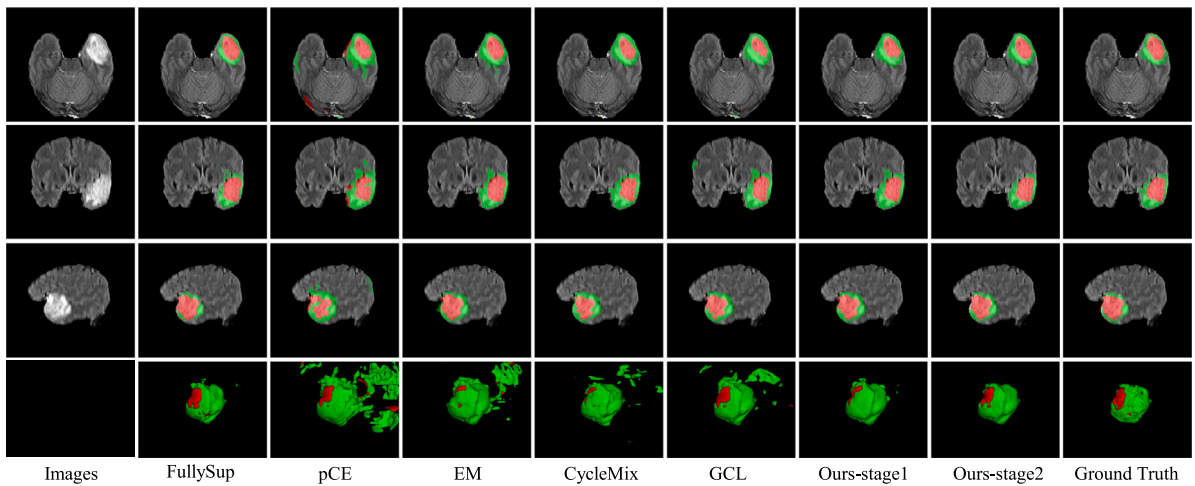


Fig. 7. Visual comparison between our method and the other weakly supervised methods on the BraTS2020 dataset. The 1st, 2nd, 3rd, and 4th rows display the visualizations of axial, coronal, sagittal, and 3D views, respectively. Best viewed in color.

Table 2

Quantitative comparison between our method DMSPS and existing weakly supervised methods on WORD testing set. The best values are highlighted in bold.

Metric	Method	Liver	Spleen	Kidney(L)	Kidney(R)	Stomach	Gallbladder	Pancreas	Mean
$DSC(\%) \uparrow$	FullySup (upper bound)	96.37±0.74	95.42±1.55	94.95±1.58	95.33±1.34	90.08±4.42	75.33±13.21	80.90±7.67	89.77±4.36
	pCE (lower bound)	86.53±3.61	86.81±5.75	86.25±4.46	89.41±2.97	61.09±12.20	56.61±20.12	61.55±10.45	75.46±8.51
	USTM (Liu et al., 2022)	80.57±3.87	86.49±7.71	87.15±4.21	89.26±2.35	77.33±6.19	63.94±17.17	67.24±9.49	78.85±6.86
	S2L (Lee and Jeong, 2020)	68.42±3.51	89.98±5.51	88.10±4.05	89.90±4.05	81.39±6.07	60.99±19.55	69.98±8.49	78.39±7.32
	EM (Grandvalet and Bengio, 2004)	89.60±1.72	87.76±5.65	87.29±3.88	81.32±4.79	77.74±7.34	65.83±16.80	73.52±7.54	80.44±6.82
	CycleMix (Zhang and Zhuang, 2022)	94.20±0.94	87.29±5.78	90.66±1.80	90.90±3.76	66.02±18.17	82.32±5.78	73.85±7.43	83.61±6.24
	GCL (Obukhov et al., 2019)	93.54±1.05	93.29±2.52	90.67±3.16	92.51±3.25	83.51±4.58	69.61±15.02	72.68±8.57	85.12±5.45
	DMSPS-stage1	94.40±0.91*	92.24±2.68	92.37±2.37*	92.96±1.73	87.91±3.79*	71.14±15.37	76.49±7.85*	86.79±4.84*
	DMSPS-stage2	94.76±0.74*	93.43±2.06	93.33±1.99*	93.56±1.46	88.41±3.66*	72.33±14.85	77.12±7.30*	87.56±4.58*
$HD_{95}(mm) \downarrow$	FullySup (upper bound)	3.45±1.40	3.01±3.31	2.84±1.05	2.47±0.71	10.01±8.36	10.50±15.80	7.71±6.86	5.71±5.36
	pCE (lower bound)	64.97±33.39	59.65±46.21	63.51±46.81	56.12±50.34	130.05±30.53	69.23±36.97	99.41±13.35	77.56±36.80
	USTM (Liu et al., 2022)	99.07±28.83	13.89±18.13	30.58±6.15	40.81±49.34	90.49±44.52	61.75±52.17	93.42±14.41	61.43±34.79
	S2L (Lee and Jeong, 2020)	48.40±14.44	6.70±6.04	6.49±1.94	6.02±2.43	38.81±41.89	26.52±29.29	34.62±29.01	23.94±17.86
	EM (Grandvalet and Bengio, 2004)	32.27±24.35	24.79±39.25	7.32±3.51	121.68±10.83	107.03±54.52	29.62±45.69	33.57±31.29	50.90±29.92
	CycleMix (Zhang and Zhuang, 2022)	6.31±5.18	12.64±13.04	4.92±3.12	4.23±3.09	16.28±13.92	17.92±14.38	10.35±7.16	10.40±8.56
	GCL (Obukhov et al., 2019)	7.29±3.13	7.24±18.11	7.31±2.21	4.39±2.00	14.58±8.55	11.79±16.43	16.66±23.68	9.90±10.59
	DMSPS-stage1	5.57±2.62*	9.42±2.79	4.42±2.03*	3.58±1.11*	9.99±6.26*	17.13±22.89	11.99±16.28	8.85±11.30
	DMSPS-stage2	4.99±1.75*	3.49±1.64	3.70±1.63*	3.28±0.84*	9.95±7.11*	9.04±13.84	9.46±7.69	6.27±4.93*

* Denotes significant improvement (p -value < 0.05 based on paired t -test) from the best existing method GCL (Obukhov et al., 2019).**Table 3**

Quantitative comparison between our method DMSPS and existing weakly supervised methods on BraTS2020 testing set. The best values are highlighted in bold.

Method	Tumor		Edema		Mean	
	$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$	$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$	$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$
FS (upper bound)	81.99 ± 21.46	9.11 ± 20.21	78.62 ± 13.46	6.51 ± 13.37	80.30 ± 17.46	7.81 ± 16.79
pCE (lower bound)	56.76 ± 24.02	74.09 ± 36.79	50.46 ± 18.55	67.87 ± 20.55	53.61 ± 21.28	70.98 ± 28.67
USTM (Liu et al., 2022)	63.90 ± 24.23	69.17 ± 44.71	46.88 ± 18.07	84.11 ± 17.65	55.39 ± 21.15	76.64 ± 31.18
S2L (Lee and Jeong, 2020)	75.80 ± 22.52	17.83 ± 28.75	61.50 ± 12.15	25.28 ± 27.08	68.65 ± 17.33	21.55 ± 27.92
EM (Grandvalet and Bengio, 2004)	70.98 ± 23.60	47.03 ± 40.45	65.67 ± 14.43	43.73 ± 34.33	68.33 ± 19.01	45.38 ± 37.39
CycleMix (Zhang and Zhuang, 2022)	77.67 ± 17.66	28.82 ± 34.24	64.24 ± 14.35	40.53 ± 28.69	70.96 ± 16.00	34.67 ± 31.47
GCL (Obukhov et al., 2019)	77.24 ± 21.90	18.28 ± 29.65	67.87 ± 12.74	29.77 ± 33.73	72.55 ± 17.32	24.02 ± 31.69
DMSPS-stage1	78.92 ± 21.75*	10.81 ± 20.61*	69.66 ± 12.43*	10.68 ± 17.68*	74.28 ± 17.09*	10.75 ± 19.15*
DMSPS-stage2	79.39 ± 21.19*	8.13 ± 16.11*	73.66 ± 12.04*	7.57 ± 13.06*	76.53 ± 16.62*	7.85 ± 14.59*

* Denotes significant improvement (p -value < 0.05 based on paired t -test) from the best existing method GCL (Obukhov et al., 2019).**Table 4**

Ablation study of our method on ACDC dataset. “Baseline (Lower bound)” denotes the UNet trained with pCE only. “FullySup (Upper bound)” represents the UNet trained with full annotations. CR stands for consistency regularization between the main and auxiliary decoders. CPS refers to the cross pseudo supervision strategy. “ $\alpha = 0.5$ ” means using the average of pseudo-labels from two branches. “no detach” means the dynamically mixed pseudo-labels in Eq. (2) are not detached. “hard” means using hard pseudo-labels as in our previous work (Luo et al., 2022a). “M” refers to the binary mask in Eq. (6) for obtaining the expanded scribble annotations. “w/o M” means using the predictions from the first-stage model for fully supervised training. The best values are highlighted in bold.

Method	RV		Myo		LV		Mean	
	$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$	$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$	$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$	$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$
Baseline (Lower bound)	46.53±14.15	194.93±31.71	46.19±11.17	171.28±20.91	58.65±20.00	169.98±21.44	50.46±15.11	178.73±24.68
DB-Net	55.20±13.48	177.55±34.23	58.26±9.10	171.84±18.74	65.86±18.36	182.57±22.26	59.77 ± 13.65	177.32±25.08
DB-Net + CR	87.63±7.10	5.19±3.97	83.48±6.30	16.18±35.66	92.03±7.24	10.49±31.63	87.71±6.88	10.62±23.76
DB-Net + CPS	87.27±6.95	8.12±12.58	83.14±6.13	20.95±34.50	90.93±8.40	18.01±37.71	87.11±7.16	15.69±28.26
Ours ($\alpha = 0.5$)	88.03±7.43	5.04±3.63	83.09±6.71	4.42±3.24	92.31±7.31	5.10±7.00	87.81±7.15	4.85±4.62
Ours (no detach)	88.01±7.57	5.16±3.84	85.38±5.50	3.81±3.76	92.21±7.14	4.45±6.36	88.53±6.74	4.47±4.65
Ours (hard)	88.04±6.39	5.10±3.62	83.51±6.10	13.79±30.22	91.93±7.16	13.63±33.85	87.83±6.55	10.84±22.56
Ours	88.26±6.76	5.09±3.74	85.62±5.40	3.78±3.72	92.95±5.84	3.78±5.35	88.94±6.00	4.22±4.27
Ours-stage2 (w/o M)	88.51±7.02	4.89±3.77	85.54±5.65	3.55±2.99	93.02±6.08	3.71±5.41	89.02±6.26	4.05±4.06
Ours-stage2	88.58±6.52	4.62±3.70	86.74±4.62	2.96±2.40	93.22±5.77	3.57±4.76	89.51±5.64	3.72±3.62
FullySup (Upper bound)	88.71±7.91	5.10±4.29	88.95±4.30	3.17±3.10	93.27±7.59	3.68±4.75	90.31±6.60	3.98±4.05

4.4. Ablation study

4.4.1. Supervision strategy for DB-Net

We used both the ACDC dataset with raw scribble annotations and the WORD dataset for ablation study. The experimental results are shown in Tables 4 and 5 respectively. We first investigated the performance difference between DB-Net and basic UNet under sole supervision of pCE loss. The results on both two datasets show that DB-Net outperformed the UNet when trained from scribbles with pCE loss only, demonstrating the effectiveness of using two decoders for improving the model’s feature learning ability. However, it was still inferior to other methods that leverage unlabeled pixels for training,

indicating the significance of incorporating unsupervised information for optimizing a model’s performance.

To investigate the effectiveness of our dynamically mixed soft pseudo labels, we compared it with two other supervision strategies based on the dual-branch network with only one stage: (1) Consistency Regularization (CR) (Ouali et al., 2020; Liu et al., 2023) that directly encourages inter-branch consistency; (2) Cross Pseudo Supervision (CPS) (Chen et al., 2021b), where each branch is supervised by hard pseudo-labels obtained by the other. Table 4 shows that on the ACDC dataset, the average DSC achieved by CR and CPS strategies was 87.71% and 87.11%, respectively, lower than the 88.94% obtained by our strategy. Similarly, Table 5 shows that on the WORD dataset, our strategy also outperforms CR and CPS, with an average DSC of

Table 5

Ablation study on WORD dataset of different supervision strategies for the dual-branch network. “Baseline (Lower bound)” denotes the 3D UNet trained with pCE only. “FullySup (Upper bound)” represents the 3D UNet trained with full annotations as upper bound. CR stands for consistency regularization between the main and auxiliary decoders. CPS refers to the cross pseudo supervision strategy. “ $\alpha = 0.5$ ” means using the average of pseudo-labels from two branches. “no detach” means the dynamically mixed pseudo-labels in Eq. (2) are not detached. “hard” means using hard pseudo-labels as in our previous work (Luo et al., 2022a). “ M ” refers to the binary mask in Eq (6) for obtaining the expanded scribble annotations. “w/o M ” means using the predictions from the first-stage model for fully supervised training. The best values are highlighted in bold.

Metric	Method	Liver	Spleen	Kidney(L)	Kidney(R)	Stomach	Gallbladder	Pancreas	Mean
$DSC(\%) \uparrow$	Baseline (Lower bound)	86.53 $\pm$ 3.61	86.81 $\pm$ 5.75	86.25 $\pm$ 4.46	89.41 $\pm$ 2.97	61.09 $\pm$ 12.20	56.61 $\pm$ 20.12	61.55 $\pm$ 10.45	75.46 $\pm$ 8.51
	DB-Net	89.57 $\pm$ 1.90	86.95 $\pm$ 5.43	85.01 $\pm$ 4.40	87.81 $\pm$ 3.07	77.17 $\pm$ 8.34	55.56 $\pm$ 18.60	70.60 $\pm$ 8.65	78.96 $\pm$ 7.20
	DB-Net + CR	92.49 $\pm$ 1.28	90.40 $\pm$ 3.41	90.26 $\pm$ 4.77	91.10 $\pm$ 2.22	86.14 $\pm$ 3.98	69.58 $\pm$ 14.86	75.24 $\pm$ 8.00	85.03 $\pm$ 5.50
	DB-Net + CPS	93.24 $\pm$ 1.48	89.80 $\pm$ 3.78	90.38 $\pm$ 2.64	91.14 $\pm$ 2.03	85.20 $\pm$ 6.57	68.84 $\pm$ 15.61	75.01 $\pm$ 7.55	84.80 $\pm$ 5.66
	Ours ($\alpha = 0.5$)	92.45 $\pm$ 1.18	92.76 $\pm$ 2.30	91.98 $\pm$ 2.00	92.99 $\pm$ 1.99	85.64 $\pm$ 4.11	66.65 $\pm$ 16.89	77.14 $\pm$ 7.78	85.66 $\pm$ 5.18
	Ours (no detach)	94.63 $\pm$ 0.82	92.67 $\pm$ 2.92	92.98 $\pm$ 2.11	92.92 $\pm$ 1.65	87.75 $\pm$ 3.80	68.69 $\pm$ 18.61	76.33 $\pm$ 7.95	86.57 $\pm$ 5.41
	Ours (hard)	88.89 $\pm$ 2.54	89.19 $\pm$ 3.93	90.14 $\pm$ 2.98	90.93 $\pm$ 2.15	77.06 $\pm$ 7.39	70.25 $\pm$ 15.57	75.02 $\pm$ 7.53	83.07 $\pm$ 6.01
	Ours	94.40 $\pm$ 0.91	92.24 $\pm$ 2.68	92.37 $\pm$ 2.37	92.96 $\pm$ 1.73	87.99 $\pm$ 3.79	71.14 $\pm$ 15.37	76.49 $\pm$ 7.85	86.79 $\pm$ 4.84
	Ours-stage2 (w/o M)	94.55 $\pm$ 0.70	92.68 $\pm$ 2.38	92.80 $\pm$ 2.10	93.09 $\pm$ 1.66	88.17 $\pm$ 4.06	70.90 $\pm$ 16.76	76.98 $\pm$ 7.12	87.02 $\pm$ 4.97
	Ours-stage2	94.76 $\pm$ 0.74	93.43 $\pm$ 2.06	93.33 $\pm$ 1.99	93.56 $\pm$ 1.46	88.41 $\pm$ 3.66	72.33 $\pm$ 14.85	77.12 $\pm$ 7.30	87.56 $\pm$ 4.58
$HD_{95}(mm) \downarrow$	FullySup (Upper bound)	96.37 $\pm$ 0.74	95.42 $\pm$ 1.55	94.95 $\pm$ 1.58	95.33 $\pm$ 1.34	90.08 $\pm$ 4.42	75.33 $\pm$ 13.21	80.90 $\pm$ 7.67	89.77 $\pm$ 4.36
	Baseline (Lower bound)	64.97 $\pm$ 33.39	59.65 $\pm$ 46.21	63.51 $\pm$ 46.81	56.12 $\pm$ 50.34	130.05 $\pm$ 30.53	69.23 $\pm$ 36.97	99.41 $\pm$ 13.35	77.56 $\pm$ 36.80
	DBNet	31.20 $\pm$ 42.20	60.41 $\pm$ 44.60	50.50 $\pm$ 46.96	8.62 $\pm$ 10.09	95.85 $\pm$ 41.86	100.86 $\pm$ 30.86	75.76 $\pm$ 31.57	60.46 $\pm$ 35.45
	DB-Net + CR	8.54 $\pm$ 8.88	17.43 $\pm$ 41.13	13.69 $\pm$ 47.25	4.65 $\pm$ 2.14	18.82 $\pm$ 37.16	21.58 $\pm$ 27.67	18.54 $\pm$ 31.42	14.75 $\pm$ 27.95
	DB-Net + CPS	13.78 $\pm$ 36.82	11.47 $\pm$ 24.12	13.50 $\pm$ 28.16	4.11 $\pm$ 0.94	25.09 $\pm$ 44.75	19.12 $\pm$ 21.54	20.98 $\pm$ 33.28	15.44 $\pm$ 27.09
	Ours ($\alpha = 0.5$)	7.88 $\pm$ 3.79	4.04 $\pm$ 2.17	5.17 $\pm$ 1.95	7.32 $\pm$ 17.83	10.94 $\pm$ 5.24	16.70 $\pm$ 17.66	9.65 $\pm$ 9.75	8.81 $\pm$ 8.34
	Ours (no detach)	4.83 $\pm$ 1.33	4.46 $\pm$ 3.27	5.19 $\pm$ 3.22	3.91 $\pm$ 1.20	17.44 $\pm$ 32.13	13.57 $\pm$ 16.81	14.39 $\pm$ 25.67	9.11 $\pm$ 11.95
	Ours (hard)	27.16 $\pm$ 30.80	11.82 $\pm$ 22.66	5.13 $\pm$ 1.74	4.87 $\pm$ 1.97	89.04 $\pm$ 42.58	16.75 $\pm$ 19.65	13.20 $\pm$ 21.18	24.00 $\pm$ 20.08
	Ours	5.57 $\pm$ 2.62	9.42 $\pm$ 2.79	4.42 $\pm$ 2.03	3.58 $\pm$ 1.11	9.99 $\pm$ 6.26	17.13 $\pm$ 22.89	11.99 $\pm$ 16.28	8.85 $\pm$ 11.30
	Ours-stage2 (w/o M)	4.87 $\pm$ 1.08	3.61 $\pm$ 1.41	3.89 $\pm$ 1.62	3.41 $\pm$ 0.95	15.12 $\pm$ 32.35	11.90 $\pm$ 15.60	9.24 $\pm$ 7.60	7.44 $\pm$ 8.66
$HD_{95}(mm) \downarrow$	Ours-stage2	4.99 $\pm$ 1.75	3.49 $\pm$ 1.64	3.70 $\pm$ 1.63	3.28 $\pm$ 0.84	9.95 $\pm$ 7.11	9.04 $\pm$ 13.84	9.46 $\pm$ 7.69	6.27 $\pm$ 4.93
	FullySup (Upper bound)	3.45 $\pm$ 1.40	3.01 $\pm$ 3.31	2.84 $\pm$ 1.05	2.47 $\pm$ 0.71	10.01 $\pm$ 8.36	10.50 $\pm$ 15.80	7.71 $\pm$ 6.86	5.71 $\pm$ 5.36

86.79% compared to 85.03% by CR and 84.80% by CPS . Furthermore, the HD_{95} scores of our method were much lower than those of CR and CPS on both datasets. These results demonstrate the effectiveness of our dynamically mixed soft pseudo-label strategy.

Then, we explored the impact of different strategies for obtaining mixed pseudo-labels: (1) “ $\alpha = 0.5$ ” that means p_1 and p_2 are simply averaged rather than dynamically mixed to obtain the pseudo-label $\hat{p}$; (2) “no detach” that means the randomly mixed soft pseudo-labels $\hat{p}$ in Eq. (2) are not detached; (3) “hard” that means replacing $\hat{p}$ in Eq. (2) with a hard pseudo-label (argmax of $\hat{p}$) as in our previous work (Luo et al., 2022a). Table 4 shows that on the ACDC dataset, setting α to a fixed value of 0.5 or using hard pseudo-labels decreased the average DSC by 1.13 and 1.11 percentage points, respectively. The average HD_{95} of using hard pseudo-labels was much larger than that of ours (10.84 mm vs 4.22 mm) according to Table 4. As shown in Table 5, on the WORD dataset, using the averaged pseudo-labels was inferior to dynamically mixed pseudo labels. Replacing “Ours” by “Ours (hard)” (Luo et al., 2022a) also led to a significant drop. The performance of our method slightly dropped when $\hat{p}$ was not detached for pseudo-label supervision for both the ACDC and WORD datasets, as supported by the results from Tables 4 and 5, showing the benefits of maintaining the independence of each branch in learning. The results demonstrate that our method outperformed the other variants of pseudo-label supervision in the first stage, showing the superiority of dynamically mixed soft pseudo-labels.

4.4.2. Effectiveness of second-stage training

Tables 4 and 5 show that for both ACDC and WORD datasets, our method in the second stage improved the performance from the first-stage model, due to the expanded annotation regions. The improvement was more obvious for small or non-convex organs. For example, in Table 4, the DSC of Myo was improved from 85.62% to 86.74%, and the corresponding HD_{95} was reduced from 3.78 mm to 2.96 mm. As shown in Table 5, the HD_{95} was reduced from 9.42 mm to 3.49 mm for spleen, and from 17.13 mm to 10.81 mm for gallbladder, respectively. In addition, we investigated using the first-stage predictions after retaining only the largest connected component on the training images for supervised training of the second-stage segmentation model, without using the confidence mask to select reliable expanded annotations, denoted as Ours-stage2 (w/o M). The results in both Tables 4 and 5 show this approach was inferior to our method using the confident mask M in the second stage.

Fig. 8 shows a visualization of the expanded annotations based on confident predictions from the DB-Net after the first stage. Note

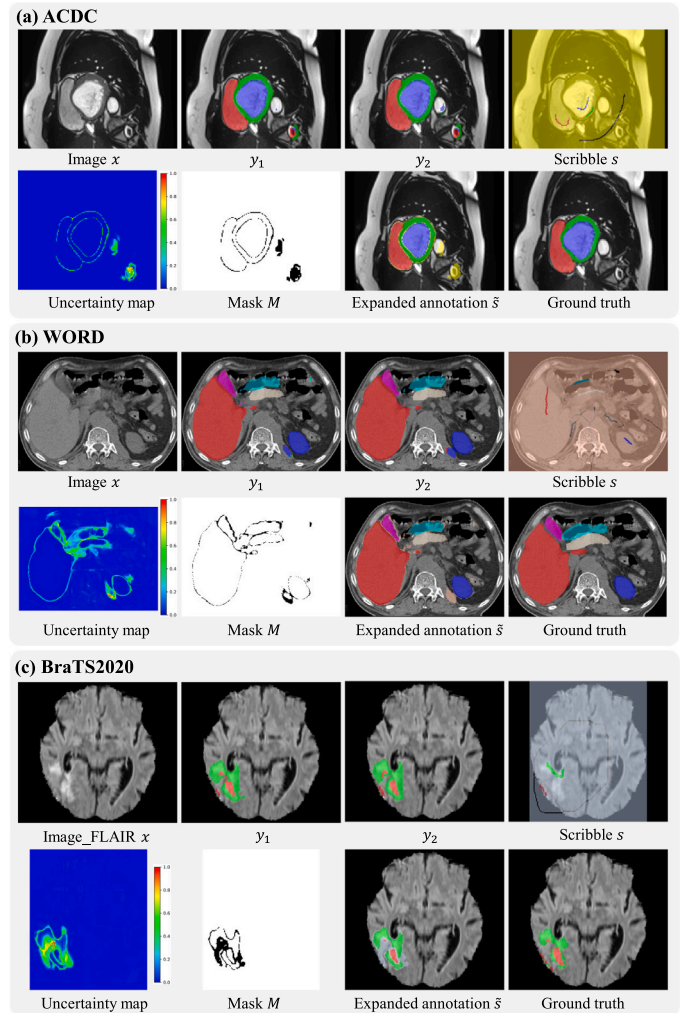


Fig. 8. Illustration of expanded annotations based on high-confident predictions from the first-stage model. Note that the unlabeled pixels in s and $\bar{s}$ are denoted by yellow color in (a), by brown color in (b) and by gray in (c), respectively. Best viewed in color.

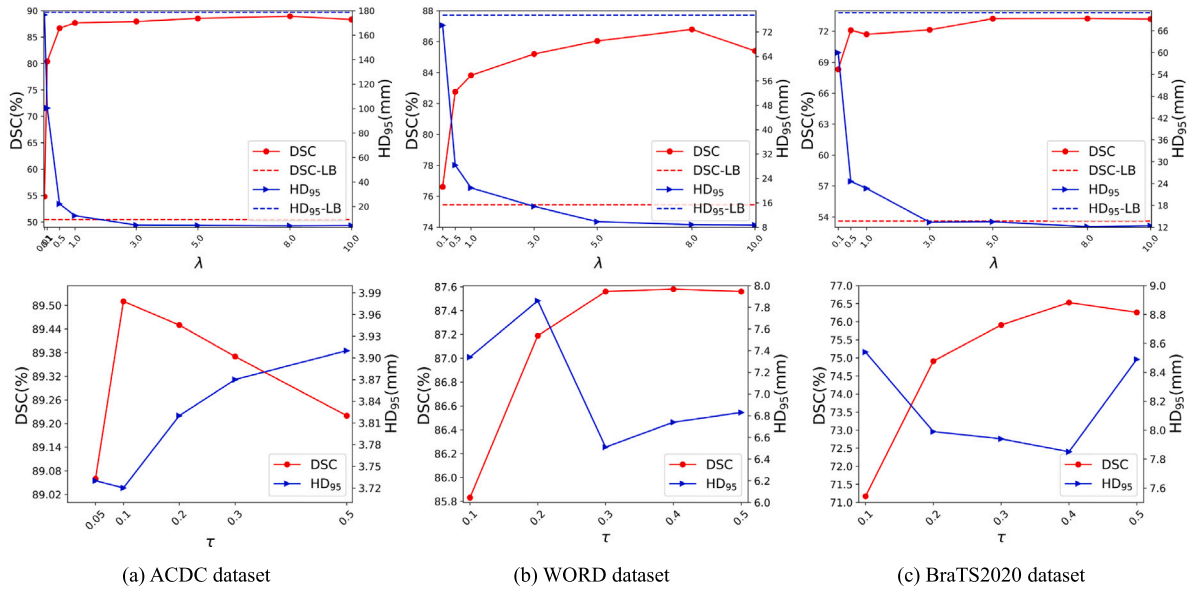


Fig. 9. Sensitivity analysis of hyper-parameter λ and τ on ACDC, WORD, BraTS2020 datasets, respectively. LB means lower bound. Best viewed in color.

Table 6

Comparison of different loss functions for soft pseudo labels on the ACDC and WORD datasets. The results were measured in the first stage of our method. The best values are highlighted in bold.

Loss function	Output	ACDC		WORD	
		$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$	$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$
MSE	θ_{d1}	86.86 $\pm$ 6.73	15.77 $\pm$ 29.62	84.23 $\pm$ 5.59	15.73 $\pm$ 28.06
	θ_{d2}	86.02 $\pm$ 7.27	23.51 $\pm$ 31.08	83.62 $\pm$ 5.81	17.21 $\pm$ 30.44
Dice	θ_{d1}	87.97 $\pm$ 6.64	6.03 $\pm$ 9.53	77.23 $\pm$ 8.19	55.62 $\pm$ 30.64
	θ_{d2}	87.87 $\pm$ 7.07	5.00 $\pm$ 6.58	77.09 $\pm$ 8.22	60.10 $\pm$ 33.00
KL	θ_{d1}	88.69 $\pm$ 6.32	4.04 $\pm$ 3.73	84.05 $\pm$ 5.84	19.70 $\pm$ 15.94
	θ_{d2}	88.38 $\pm$ 6.60	4.31 $\pm$ 4.02	83.59 $\pm$ 5.68	17.85 $\pm$ 32.52
CE	θ_{d1}	88.94 $\pm$ 6.00	4.22 $\pm$ 4.27	86.70 $\pm$ 4.84	8.85 $\pm$ 11.30
	θ_{d2}	88.78 $\pm$ 6.21	3.99 $\pm$ 3.45	86.59 $\pm$ 5.00	7.46 $\pm$ 5.89

Table 7

Comparison of computational complexity between our method DMSPS and existing weakly supervised methods on ACDC segmentation task using UNet backbone. Note that these methods have the same inference time (0.5 s/volume) due to the same backbone.

Method	Model parameters	Training time
FS(upper bound)	1.73 M	1.5 h
pCE(lower bound)	1.73 M	1.2 h
USTM (Liu et al., 2022)	3.46 M	6.9 h
S2L (Lee and Jeong, 2020)	1.73 M	5.3 h
EM (Grandvalet and Bengio, 2004)	1.73 M	1.9 h
CycleMix (Zhang and Zhuang, 2022)	1.73 M	11.5 h
Rloss (Obukhov et al., 2019)	1.73 M	3.0 h
DMSPS-stage1		3.0 h
DMSPS-stage2	2.33 M	2.0 h

that y_1 and y_2 are the predictions from the two branches, respectively. The uncertainty map effectively identifies uncertain regions that are near the boundary or noisy. The uncertainty-filtered expanded annotations $\tilde{s}$ effectively grow the raw scribbles s to much larger regions with high precision, and provide more supervisory information for $\mathcal{L}_{pCE}$, while the unannotated regions with soft pseudo-label loss $\mathcal{L}_{SPS}$ are much reduced, which avoids the model being affected by incorrect pseudo-labels based on the high-confidence region mask M . Therefore, the expanded annotated regions with high reliability and

smaller unannotated pixels are beneficial for improvement of model performance.

4.4.3. Hyper-parameter analysis

Note that α in Eq. (1) is not a hyper-parameter of our method, as it is randomly sampled from $U(0, 1)$ during training. Our method has only two main hyper-parameters: λ in Eq. (4) that controls the weight of $\mathcal{L}_{SPS}$, and τ in Eq. (6) that controls the threshold of uncertainty for reliable pseudo-label selection. We measured the segmentation performance of the first-stage model when λ was set to 0.01, 0.1, 0.5, 1.0, 3.0, 5.0, 8.0 and 10.0, respectively. Fig. 9 shows the evolution of average DSC and HD_{95} on the three datasets. It can be observed that on the ACDC dataset, setting λ to 0.01 does not bring much performance gain compared with not using the $\mathcal{L}_{SPS}$ term, and increasing λ from 0.1 to 1.0 leads to a large performance improvement on the three datasets. When λ changes from 1.0 to 10.0, the performance was relatively stable on the ACDC and BraTS2020 datasets. Overall, we found that the best λ for all three datasets was 8.0 in our experiments, and performance was not very sensitive to λ when it is in [1.0, 10.0].

We also investigated the performance of our method based on different values of τ when it was set in the range of (0.0, 0.5). As shown in Fig. 9, for the relatively simple target organs in the ACDC dataset, the optimal performance is achieved when τ is 0.1. For the more complex target organs and lesions in the WORD and BraTS2020 datasets, the best performance is obtained at $\tau = 0.3$ and $\tau = 0.4$, respectively. This is because that on the ACDC dataset where the organs are easier to

Table 8

Quantitative comparison between our method DMSPS and the best existing weakly supervised method GCL across different backbones under raw scribbles supervision on ACDC testing set for 2D segmentation. The best values are highlighted in bold for each backbone.

Backbone (2D)	Method	RV		Myo		LV		Mean	
		$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$	$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$	$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$	$DSC(\%) \uparrow$	$HD_{95}(mm) \downarrow$
UNet++ (CNN) (Zhou et al., 2019)	FS	88.87 $\pm$ 7.73	5.01 $\pm$ 4.24	88.06 $\pm$ 5.74	3.51 $\pm$ 4.26	93.13 $\pm$ 7.54	3.97 $\pm$ 4.72	90.02 $\pm$ 67.01	4.16 $\pm$ 4.40
	pCE	49.21 $\pm$ 14.68	193.47 $\pm$ 31.14	48.33 $\pm$ 9.45	172.56 $\pm$ 21.38	73.00 $\pm$ 15.22	142.44 $\pm$ 54.26	56.85 $\pm$ 13.11	169.49 $\pm$ 35.59
	GCL	88.00 $\pm$ 6.47	5.02 $\pm$ 3.31	84.34 $\pm$ 5.20	4.76 $\pm$ 5.86	91.42 $\pm$ 8.28	6.21 $\pm$ 8.44	87.92 $\pm$ 6.65	5.33 $\pm$ 5.87
	DMSPS-stage1	87.98 $\pm$ 7.63	4.99 $\pm$ 3.81	85.53 $\pm$ 5.47	3.73 $\pm$ 3.42	92.40 $\pm$ 8.06	4.51 $\pm$ 6.41	88.64 $\pm$ 7.05	4.41 $\pm$ 4.55
	DMSPS-stage2	88.11 $\pm$ 7.60	4.76 $\pm$ 3.95	86.92 $\pm$ 4.57	3.15 $\pm$ 2.54	92.80 $\pm$ 5.84	4.13 $\pm$ 5.26	89.28 $\pm$ 5.01	4.02 $\pm$ 3.91
SwinUNet (ViT) (Cao et al., 2022)	FS	83.76 $\pm$ 9.79	7.57 $\pm$ 4.55	81.59 $\pm$ 7.93	3.98 $\pm$ 1.99	89.57 $\pm$ 8.35	5.59 $\pm$ 5.56	84.97 $\pm$ 8.69	5.71 $\pm$ 4.04
	pCE	78.86 $\pm$ 13.34	45.83 $\pm$ 55.24	76.50 $\pm$ 7.82	18.68 $\pm$ 30.46	88.04 $\pm$ 7.41	11.58 $\pm$ 21.27	81.14 $\pm$ 9.53	25.36 $\pm$ 35.66
	GCL	80.49 $\pm$ 8.04	7.98 $\pm$ 4.01	77.98 $\pm$ 7.01	5.64 $\pm$ 2.74	87.99 $\pm$ 7.61	5.94 $\pm$ 4.98	82.15 $\pm$ 7.55	6.52 $\pm$ 3.91
	DMSPS-stage1	82.47 $\pm$ 7.81	7.14 $\pm$ 4.62	79.68 $\pm$ 6.60	4.88 $\pm$ 3.14	89.32 $\pm$ 7.72	4.80 $\pm$ 3.87	83.82 $\pm$ 7.37	5.61 $\pm$ 3.88
	DMSPS-stage2	81.90 $\pm$ 7.50	7.25 $\pm$ 4.41	80.53 $\pm$ 5.96	4.24 $\pm$ 2.13	89.54 $\pm$ 7.74	4.42 $\pm$ 3.39	84.00 $\pm$ 7.07	5.30 $\pm$ 3.31

segment, the model's predictions have a low entropy in most regions, a smaller τ value is needed to reject sufficient uncertain regions. On the WORD and BraTS2020 datasets where more classes are involved or the task is more difficult, the model tend to obtain high entropy in a larger region, and a relatively high τ value is needed to keep sufficient expanded annotations.

4.4.4. Comparison of loss functions for soft pseudo-labels

As our pseudo-labels are soft labels in $\mathcal{L}_{SPS}$, we explored the impact of utilizing different losses, including MSE, Dice, KL, and CE loss. The corresponding results of our method in the first stage on the ACDC and WORD datasets are shown in Table 6. It can be observed that for both datasets, the CE loss outperformed the others. Therefore, we finally adopted the CE loss as the soft pseudo-label loss $L(\cdot \| \cdot)$ in Eq. (2).

4.4.5. Comparison between two branches for prediction

We also compared the average performance between the two branches of DB-Net on the ACDC and WORD datasets, and the results are shown in Table 6. It can be observed that the two branches have a very subtle difference under different soft pseudo-label losses. For example, when CE loss was used, the average DSC were 88.94% and 88.78% for the two branches on the ACDC dataset, and 86.70% and 86.95% on the WORD, respectively. Though the primary branch has a slightly better DSC , the average HD_{95} in the auxiliary branch was slightly better. Due to their similar overall performance, we used the primary branch for inference.

4.5. Discussion

4.5.1. Effectiveness of soft pseudo-labels and two-stage training

The core challenge for learning from scribble annotations is that it is hard to obtain accurate pseudo-labels for the large amount of unannotated pixels. Therefore, effective uncertainty estimation and well-calibrated predictions are essential for improving the performance. In this work, our method has three important mechanisms to achieve this goal.

First, we use two prediction branches with dynamic mixture, which acts as a random ensemble of the two branches. It not only avoids the bias of a single prediction head, but also improves the diversity of pseudo-labels to reduce over-fitting.

Second, we use soft pseudo-labels rather than hard one-hot ones for supervision, which is more noise-robust. In our previous version DMPLS (Luo et al., 2022a), dynamically mixed hard pseudo-labels were used. However, hard pseudo-labels are overconfident, i.e., once they are incorrect with too much noise, it is difficult for the model to escape from being misled by the noise, especially in the early training stage under extreme sparse annotations where the pseudo-labels have a low accuracy. As shown in Fig. 4, the performance of DMPLS drops sharply when the length of the scribbles is reduced to 1/4 and even shorter. In contrast, the soft pseudo-labels have the following advantages: (1) Soft pseudo-labels can represent the degree of similarity/ambiguity between different categories, and provide supervision signals with more informative probability distributions than hard labels. (2) Soft pseudo-labels

can better alleviate the impact of erroneous pseudo-labels, because they lead to lower gradient values than hard labels in noisy regions.

Thirdly, we gradually expand the annotated pixels via the two-stage training framework, where only well calibrated and highly reliable predictions from the first-stage model are selected as expanded annotations for training the second-stage model. To obtain expanded annotations, we take the entropy of the average prediction from the two branches as uncertainty estimation. Note that it makes the predictions better calibrated than the typical entropy based on a single prediction. Specifically, if the two branches predict opposite class labels for a pixel with high confidence, the single prediction-based entropy will be low and cannot indicate the unreliability. In contrast, entropy of $\bar{p}$ will still be high in that case, and be able to detect such unreliable predictions due to the two branches' inconsistency. With the expanded annotations, the second-stage model has a lower risk of over-fitting, leading to higher quality of pseudo-labels for unannotated pixels, and the predictions could be better calibrated than directly learning from the raw sparse annotations.

4.5.2. Computational analysis

Due to the use of auxiliary decoder, our method has a higher time and memory consumption than single decoder-based methods. The auxiliary decoder only leads to a moderate increase of model size, such as from 1.73M to 2.33M when using UNet as the backbone, as shown in Table 7. Training the dual-branch network in the first stage takes about twice the training time for UNet + pCE. The training time of our first-stage model was approximately 3.0, 10.0 and 1.0 for the ACDC, WORD and BraTS2020 datasets, respectively. Note the second-stage model can be initialized by the first-stage model, which saves time for training when combined with early-stopping. The training time on the three datasets for the second stage was around 2.0, 6.0 and 3.5 h, respectively.

For inference on testing images, our method ignores the auxiliary decoder, and only utilizes the encoder and main decoder for segmentation, so the inference time is nearly equivalent to that of a standard single-decoder UNet architecture. On the ACDC dataset, the inference time for a volume was around 0.5 s, and the memory usage was around 1.25 GB. The inference time on the WORD and BraTS2020 datasets was 8.0 and 3.0 s per 3D volume, respectively, and the corresponding memory usage was 2.4 GB and 1.9 GB, respectively.

4.5.3. Generalizability to different backbones

We further investigated the generalizability of our method across different segmentation backbones, including CNN-based and Vision Transformer (ViT)-based neural networks. Note that other than the network structure, the hyper-parameters were set as the same as those described in Section 4.2.

On the ACDC dataset with the original scribbles by Valvano et al. (2021), we replaced UNet (Ronneberger et al., 2015) by CNN-based UNet++ (Zhou et al., 2019) and ViT-based SwinUNet (Cao et al., 2022), respectively. Our method was compared with the pCE (baseline) and GCL (Obukhov et al., 2019) that achieved the best performance among existing methods in Tables 1–3. As shown in Table 8, DMSPS significantly outperformed GCL in terms of DSC when using the UNet++

Table 9

Quantitative comparison between our method DMSPS and the best existing weakly supervised method GCL across different backbones on WORD testing set for 3D segmentation. The best values are highlighted in bold for each backbone.

Backbone (3D)	Metric	Method	Liver	Spleen	Kidney(L)	Kidney(R)	Stomach	Gallbladder	Pancreas	Mean
	$DSC(\%) \uparrow$	FS	96.26 $\pm$ 0.81	95.39 $\pm$ 1.34	94.76 $\pm$ 1.28	95.21 $\pm$ 1.32	89.90 $\pm$ 3.72	78.05 $\pm$ 12.59	81.08 $\pm$ 7.32	90.09 $\pm$ 4.05
		pCE	91.41 $\pm$ 1.26	88.28 $\pm$ 4.04	87.46 $\pm$ 4.24	87.52 $\pm$ 2.81	84.81 $\pm$ 4.72	67.39 $\pm$ 18.14	75.14 $\pm$ 7.88	83.15 $\pm$ 6.15
		GCL	94.01 $\pm$ 0.96	92.91 $\pm$ 2.67	92.81 $\pm$ 1.76	93.19 $\pm$ 2.32	84.59 $\pm$ 4.33	67.33 $\pm$ 16.51	72.91 $\pm$ 8.64	85.39 $\pm$ 5.31
		DMSPS-stage1	94.95 $\pm$ 0.77	92.75 $\pm$ 2.21	93.24 $\pm$ 1.95	93.44 $\pm$ 1.69	88.64 $\pm$ 3.62	72.23 $\pm$ 16.56	78.41 $\pm$ 6.89	87.67 $\pm$ 4.81
		DMSPS-stage2	95.19 $\pm$ 0.69	93.66 $\pm$ 1.79	93.53 $\pm$ 1.77	93.84 $\pm$ 1.61	88.98 $\pm$ 3.57	73.92 $\pm$ 15.48	79.04 $\pm$ 6.91	88.31 $\pm$ 4.54
Attention UNet (CNN) (Oktay et al., 2018)	$HD_{95}(mm) \downarrow$	FS	3.83 $\pm$ 2.03	2.49 $\pm$ 0.80	4.82 $\pm$ 0.92	2.70 $\pm$ 0.73	9.12 $\pm$ 6.73	9.29 $\pm$ 14.93	7.82 $\pm$ 6.81	5.72 $\pm$ 5.89
		pCE	10.32 $\pm$ 7.63	15.58 $\pm$ 33.54	8.98 $\pm$ 13.88	6.64 $\pm$ 2.10	15.40 $\pm$ 16.76	27.36 $\pm$ 25.74	9.69 $\pm$ 7.72	13.44 $\pm$ 15.20
		GCL	6.32 $\pm$ 1.19	4.37 $\pm$ 2.45	5.78 $\pm$ 2.14	10.30 $\pm$ 26.13	11.99 $\pm$ 6.69	13.13 $\pm$ 15.55	15.96 $\pm$ 22.47	9.69 $\pm$ 10.94
		DMSPS-stage1	4.67 $\pm$ 1.35	7.31 $\pm$ 1.80	4.33 $\pm$ 1.90	3.38 $\pm$ 1.01	9.97 $\pm$ 6.42	12.82 $\pm$ 15.77	8.21 $\pm$ 6.52	7.24 $\pm$ 7.28
		DMSPS-stage2	4.42 $\pm$ 0.96	3.14 $\pm$ 0.89	4.39 $\pm$ 1.97	3.33 $\pm$ 1.06	9.90 $\pm$ 6.55	10.32 $\pm$ 14.71	8.20 $\pm$ 6.56	6.25 $\pm$ 4.67
	$DSC(\%) \uparrow$	FS	96.56 $\pm$ 1.52	93.77 $\pm$ 3.42	92.99 $\pm$ 3.36	92.36 $\pm$ 4.26	85.19 $\pm$ 7.45	65.92 $\pm$ 17.15	74.03 $\pm$ 11.52	85.83 $\pm$ 6.95
		pCE	90.40 $\pm$ 1.95	87.20 $\pm$ 5.53	84.64 $\pm$ 6.51	84.17 $\pm$ 5.14	74.72 $\pm$ 8.07	58.44 $\pm$ 17.58	65.60 $\pm$ 9.29	77.88 $\pm$ 7.72
		GCL	93.23 $\pm$ 1.70	90.74 $\pm$ 5.42	89.70 $\pm$ 5.88	90.78 $\pm$ 7.44	77.28 $\pm$ 8.26	64.21 $\pm$ 18.69	68.88 $\pm$ 9.49	82.12 $\pm$ 8.13
		DMSPS-stage1	94.10 $\pm$ 1.56	89.84 $\pm$ 5.61	88.88 $\pm$ 5.14	90.32 $\pm$ 4.18	80.22 $\pm$ 6.60	65.26 $\pm$ 17.56	71.16 $\pm$ 8.80	82.83 $\pm$ 7.06
		DMSPS-stage2	94.48 $\pm$ 1.51	91.65 $\pm$ 4.06	90.17 $\pm$ 3.99	90.58 $\pm$ 4.67	79.74 $\pm$ 7.85	66.55 $\pm$ 18.81	71.02 $\pm$ 9.48	83.45 $\pm$ 7.87
UNETR (ViT) (Hatamizadeh et al., 2022)	$HD_{95}(mm) \downarrow$	FS	5.12 $\pm$ 4.14	4.98 $\pm$ 7.14	4.43 $\pm$ 4.29	6.09 $\pm$ 10.30	15.56 $\pm$ 17.34	14.26 $\pm$ 15.59	10.32 $\pm$ 8.22	8.68 $\pm$ 9.57
		pCE	17.86 $\pm$ 12.01	26.06 $\pm$ 27.83	24.44 $\pm$ 28.48	58.26 $\pm$ 44.02	66.47 $\pm$ 34.61	45.92 $\pm$ 50.60	76.57 $\pm$ 28.94	45.08 $\pm$ 32.37
		GCL	8.79 $\pm$ 6.16	7.58 $\pm$ 8.72	10.16 $\pm$ 12.40	6.42 $\pm$ 7.33	15.66 $\pm$ 7.42	18.90 $\pm$ 17.05	15.16 $\pm$ 19.24	11.81 $\pm$ 11.19
		DMSPS-stage1	7.01 $\pm$ 5.74	8.81 $\pm$ 9.98	8.69 $\pm$ 11.52	5.37 $\pm$ 1.68	16.60 $\pm$ 16.66	17.64 $\pm$ 19.99	10.96 $\pm$ 8.15	10.73 $\pm$ 10.53
		DMSPS-stage2	6.27 $\pm$ 3.24	6.22 $\pm$ 7.56	5.71 $\pm$ 2.24	5.23 $\pm$ 2.93	18.77 $\pm$ 17.77	13.30 $\pm$ 14.67	10.97 $\pm$ 7.41	9.50 $\pm$ 7.98

(89.28% vs. 87.92%) and SwinUNet (84.00% vs. 82.15%) backbones, respectively. On the WORD dataset, as shown in Table 9, with the backbone of 3D Attention UNet (Oktay et al., 2018), our method improved the average DSC from 83.15% to 88.31% compared with pCE, and it also outperformed GCL (85.39%). The same observation can be obtained with the backbone of ViT-based UNETR (Hatamizadeh et al., 2022).

5. Conclusion

In this work, we present a two-stage framework that leverages dynamically mixed soft pseudo-labels for scribble-supervised medical image segmentation. It employs a dual-branch network to learn from scribble annotations. In each stage, outputs from the two branches are dynamically mixed to generate soft pseudo-labels that are more noise-robust than hard pseudo-labels for handling unannotated pixels. To further deal with sparse annotations, the raw scribbles are expanded based on reliable predictions in the first-stage model with uncertainty-filtering, and then used as additional annotated pixels to train the second-stage model for enhancing segmentation performance. Experiments on three public datasets for cardiac structure, multiple abdominal organ and brain tumor segmentation showed the effectiveness of the proposed method, and it obtained superior performance over five existing scribble-supervised segmentation methods and different segmentation backbones, especially when the scribbles become extremely sparse. Our method shows promise for attaining high segmentation performance while substantially reducing the annotation effort required.

CRedit authorship contribution statement

Meng Han: Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Xiangde Luo:** Supervision, Methodology, Conceptualization. **Xiangjiang Xie:** Validation. **Wenjun Liao:** Resources, Data curation. **Shichuan Zhang:** Resources, Data curation. **Tao Song:** Methodology. **Guotai Wang:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Shaoting Zhang:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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